SpotNFix : Smart Pothole Detection and Reporting System

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Abstract

- SpotNFix is an AI-based platform for reporting and managing potholes using geotagged images.
- Deep learning models (YOLO and MobileNetV2) detect potholes and classify their severity for repair prioritization.
- The system connects citizens, authorities, and contractors through a transparent digital repair workflow.

Objective

- To develop an AI-based system that detects potholes and classifies their severity from uploaded images.
- To implement a transparent contractor bidding system for efficient repair allocation.
- To integrate geolocation for accurate pothole mapping and monitoring.
- To create a digital workflow connecting citizens, authorities, and contractors.

Dataset Details

Roboflow Pothole dataset for detection and a Kaggle annotated pothole dataset for severity classification.

2,771+ Detection Images

717 SEVERITY SAMPLES

3 Severity Levels

System Design

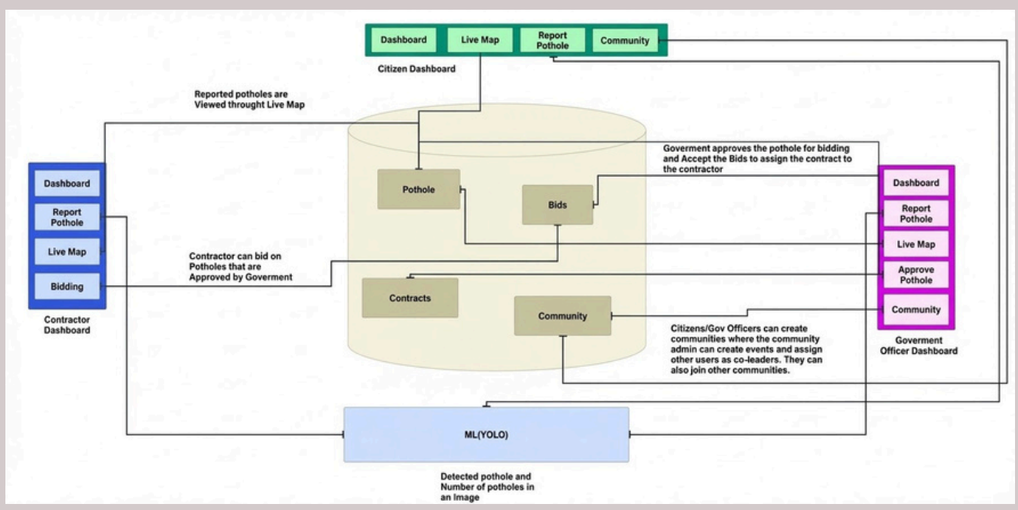

Fig. 1. System Architecture

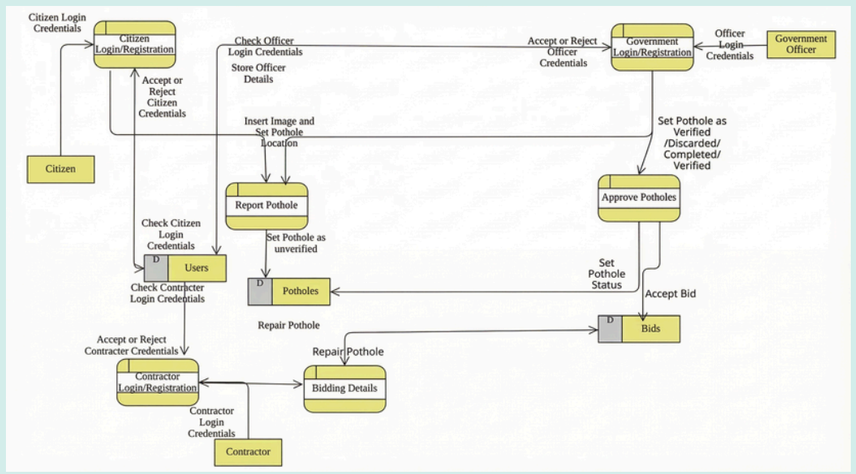

Fig.2. Dataflow Diagram

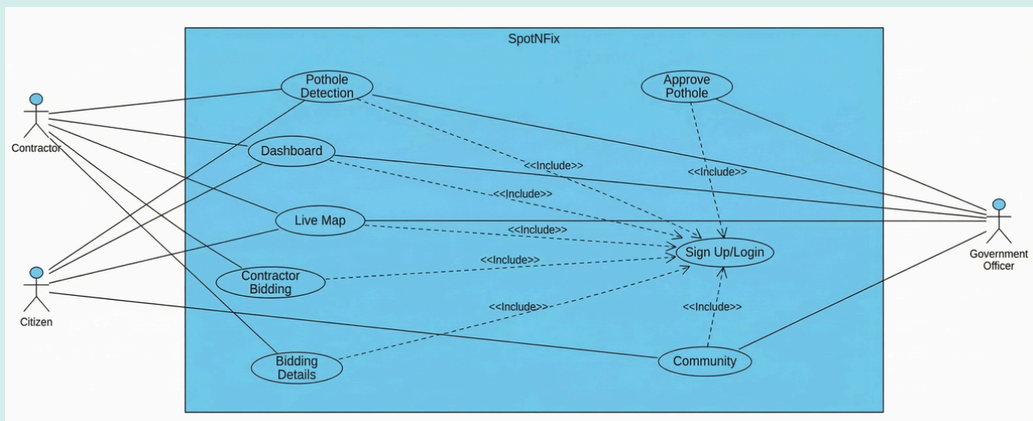

Fig.4. Usecase Diagram

Fig. 3. Tools and Technologies

Model Results

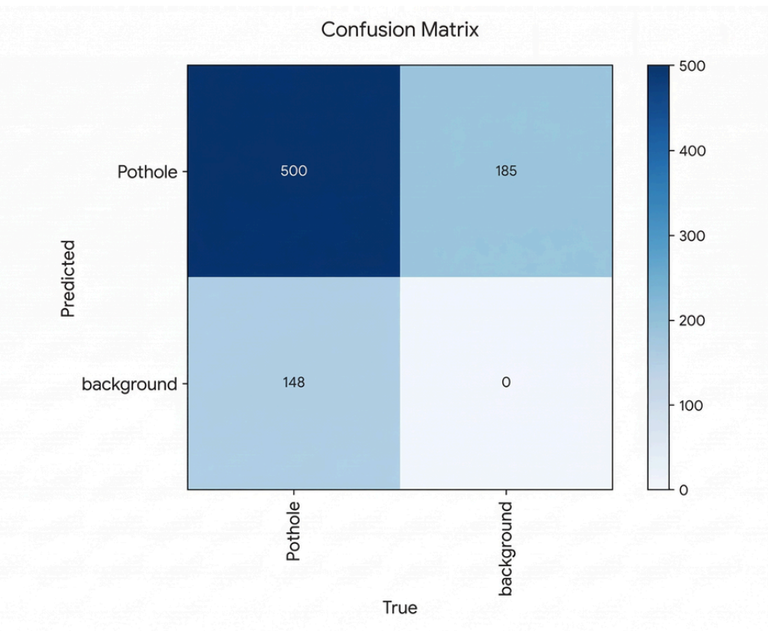

Fig. 5. YOLO Confusion Matrix

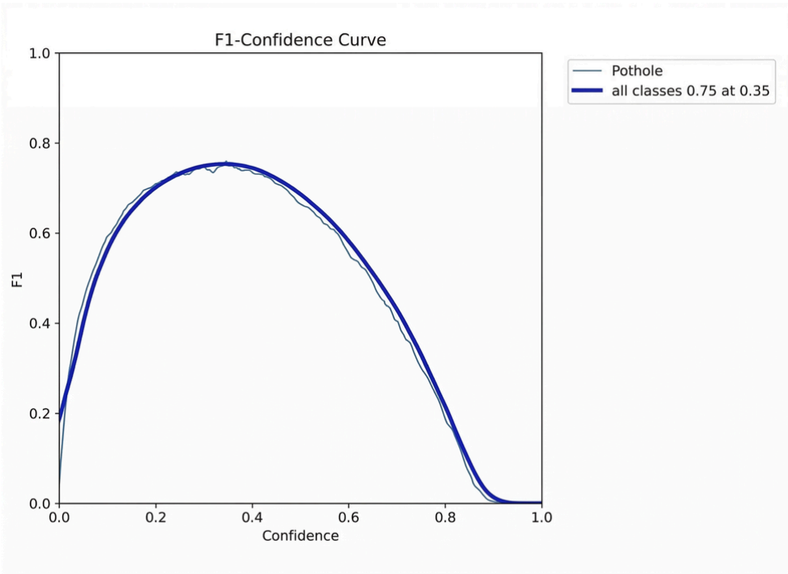

Fig. 6. F1 Confidence Curve

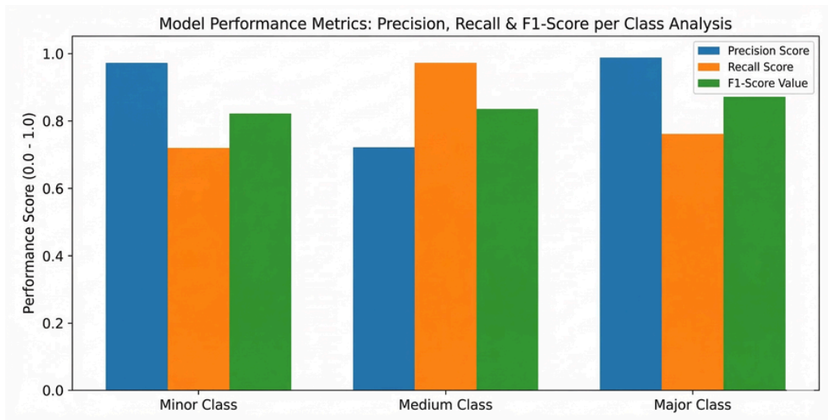

Fig. 7. Class wise Severity Metrics

SDG Goal

11 SUSTAINABLE CITIES AND COMMUNITIES

Conclusion

- SpotNFix demonstrates how AI-based pothole detection and severity classification can improve the efficiency and transparency of road maintenance systems.
- By integrating automated verification, geolocation tracking, and contractor bidding, the system enables faster and more accountable pothole repair management.

References

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2. K. R. Ahmed, "Smart pothole detection using deep learning based on dilated convolution," Sensors, vol. 21, no. 24, p. 8406, 2021.
3. J. Ren and D. Liu, "PADS: A reliable pothole detection system using machine learning," in Lecture Notes in Computer Science, vol. 10135, Springer, 2017, pp. 327–338.